Lecture 5
Stresses in Beams
(Advanced Topics)

Prof. J. J. Wang

cewang@hku.hk

Room 6-04, Haking Wong Building

Textbook: Gere & Goodno, Mechanics of Materials, Eighth Edition, SI Edition

Outline

- Composite beams
- The transformed-section method
- Beams with axial loads
- Doubly symmetric beams under inclined loads

Sections 6.2, 6.3 and 6.4 of the textbook.

What you should be able to do

- Find the neutral axis of a beam made of two materials.
- Use the transformed section to get the stresses.
- Deal with a load that is not in a plane of symmetry.
- Find the neutral axis of a doubly symmetric beam under an inclined load, and the largest stress that goes with it.

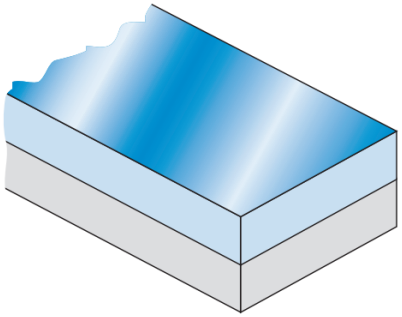
Lecture 5

Composite Beams

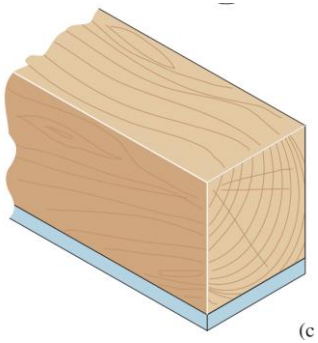
Why composite beams?

- A single material may be too flexible, too heavy, or too expensive.
- Combining two materials can give stiffness where it is needed.
- A steel plate under a timber beam adds stiffness where the tension is largest.
- The parts must be bonded so that they bend together as one beam.

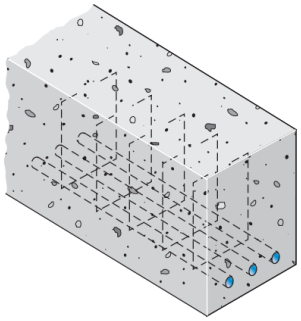
Composite beams



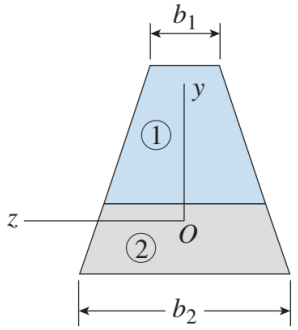
Bimetallic



Wood–steel



Concrete



Plated

The two materials are bonded, so they bend as one beam and share the same curvature.

But at the same distance y from the axis they carry different stresses, because their moduli E_1 and E_2 differ.

Strain: plane sections remain plane

A plane cross section stays plane and simply rotates about the neutral axis.

Points farther from the axis stretch or shorten more, so the strain varies linearly with y.

$$\varepsilon_x = -\frac{y}{\rho} = -\kappa\, y \qquad (1)$$

ε_x = longitudinal strain; y = distance from the neutral axis;

ρ = radius of curvature; $\kappa = 1/\rho$ = curvature.

The minus sign means that the part above the axis is shortened.

Stress: Hooke's law in each material

Each material obeys Hooke's law, stress = $E \times$ strain, so we simply multiply the strain by the modulus of that material.

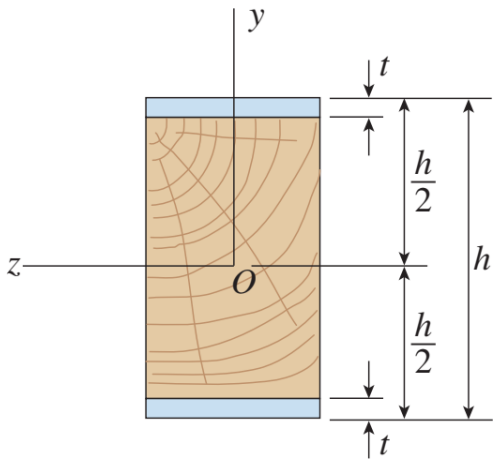
At a bonded surface the strain is the same in both, but E differs, so the stresses jump.

$$\sigma_{x1} = -E_1 \kappa\, y, \qquad \sigma_{x2} = -E_2 \kappa\, y \tag{2}$$

σ_{x1}, σ_{x2} = normal stress in material 1 and material 2 at the same point.

Where is the neutral axis?

- The neutral axis is the line where the stress is zero.
- For one material it passes through the centroid.
- For two materials it shifts towards the stiffer one.



The axis lies between the two materials

We cannot guess its position, so we find it from equilibrium.

Neutral axis — step 1: no axial force

Assume the axis is the neutral axis, so the two stress formulas hold with *y* measured from it.

A beam in pure bending carries no axial force, so the stresses must cancel out over the whole cross section.

$$\int_{A_1} \sigma_{x1}\, dA + \int_{A_2} \sigma_{x2}\, dA = 0 \qquad (3)$$

dA is a small element of area; A_1 and A_2 are the two parts of the section.

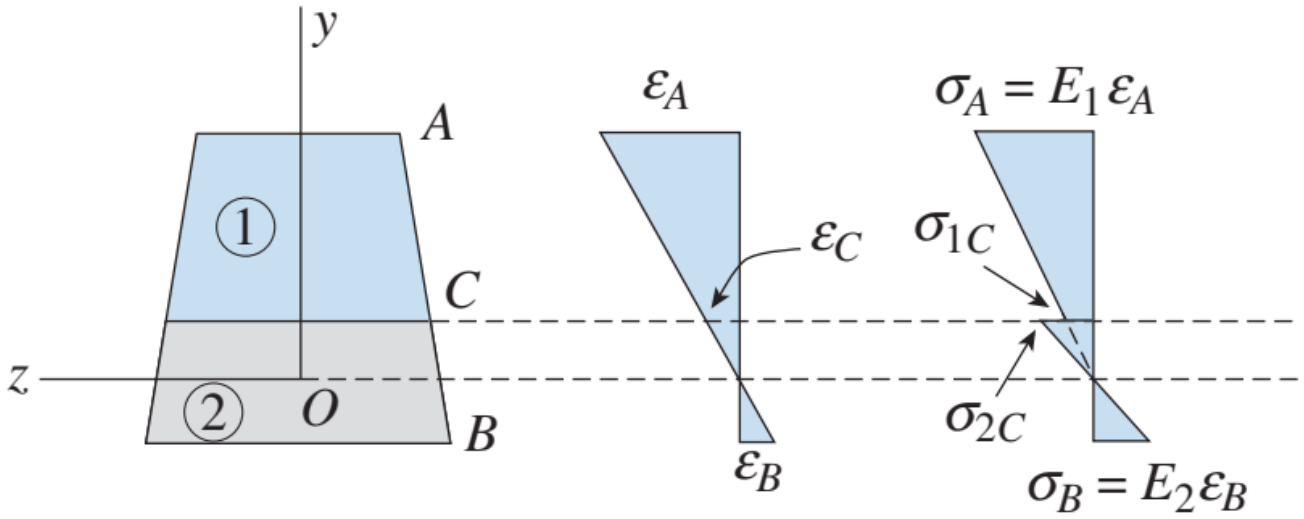
Neutral axis — step 2: the equation

Substitute the two stresses. The curvature is the same at every point of the section, so it can be taken outside the integrals and cancelled.

$$E_1 \int_{A_1} y \, dA + E_2 \int_{A_2} y \, dA = 0 \qquad (4)$$

This is the equation that locates the neutral axis of a composite beam.

Neutral axis — the meaning



Each integral is the first moment of one part of the area about the axis.

The equation is a weighted balance: the stiff part pulls the axis towards itself, so the axis does not pass through the geometric centroid.

The bending moment carried by the section

The stresses also form the applied bending moment. Taking moments about the neutral axis, both materials contribute:

$$M = \int_{A_1} \sigma_{x1}\, y\, dA + \int_{A_2} \sigma_{x2}\, y\, dA \qquad (5)$$

Put $\sigma_{x1} = -E_1 \kappa y$ and $\sigma_{x2} = -E_2 \kappa y$ in and take the constants out:

$$M = E_1 \kappa \int_{A_1} y^2\, dA + E_2 \kappa \int_{A_2} y^2\, dA$$

The moment-curvature relationship

Each integral is the second moment of area of one part about the neutral axis. Give them their usual names:

$$I_1 = \int_{A_1} y^2 \, dA, \qquad I_2 = \int_{A_2} y^2 \, dA$$

$$M = \kappa \, (E_1 \, I_1 + E_2 \, I_2)$$

$$\kappa = \frac{1}{\rho} = \frac{M}{E_1 \, I_1 + E_2 \, I_2} \tag{6}$$

The denominator $E_1 I_1 + E_2 I_2$ is the flexural rigidity. For one material it reduces to EI.

The flexure formulas for a composite beam

Put the curvature back into $\sigma = -E\kappa y$. Material 1 uses E_1, material 2 uses E_2, and both share the same denominator:

$$\sigma_{x1} = -\frac{M\, y\, E_1}{E_1\, I_1 + E_2\, I_2}, \qquad \sigma_{x2} = -\frac{M\, y\, E_2}{E_1\, I_1 + E_2\, I_2} \tag{7}$$

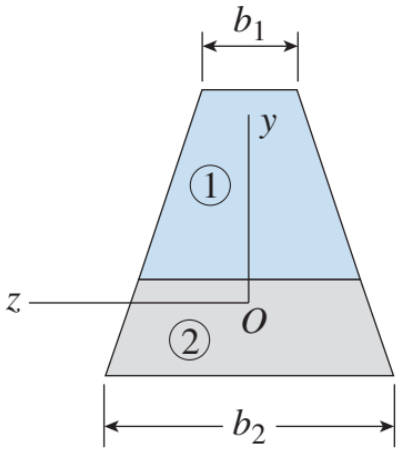
Read them together and the whole method appears: the strain is the same on both sides of a bonded surface, so the stresses there are in the ratio E_2/E_1.

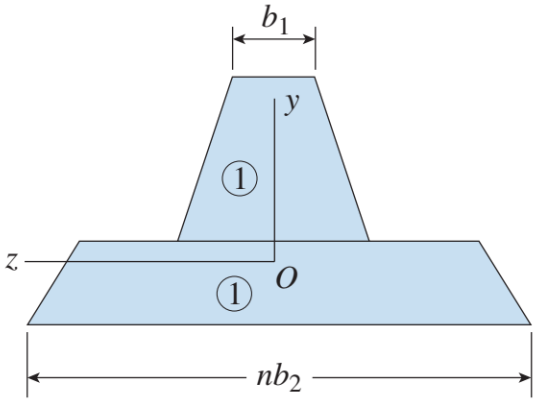
If $E_1 = E_2 = E$ the two formulas collapse to the single-material $\sigma = -M\, y\, /\, I$.

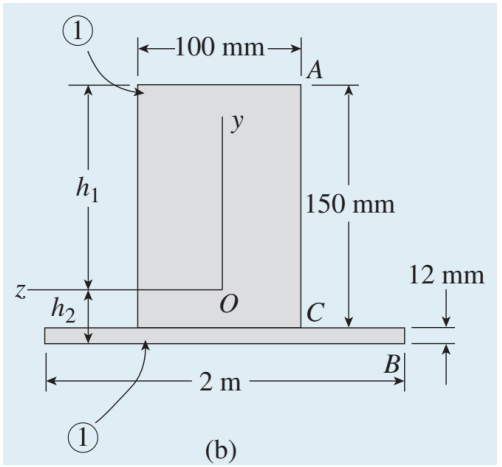
Lecture 5

Transformed-Section Method

The transformed section: the idea







- Imagine replacing the two materials by material 1 only.
- To carry the same force, widen the part of material 2 in proportion to its stiffness.
- The widened section is called the transformed section.

Why the width is multiplied by *n*

A strip of material 2 is *n* times stiffer than the same strip of material 1.

Making it *n* times wider makes it carry *n* times the force at the same strain, so nothing changes.

$$n = \frac{E_2}{E_1}, \qquad b_2 \rightarrow n\, b_2 \tag{8}$$

n = modular ratio; b_2 = width of the part made of material 2. The transformed beam then bends like an ordinary beam of one material.

Stresses in the transformed beam

The transformed section is one material, so it has a single second moment about the neutral axis. Widening part 2 by *n* multiplies its second moment by *n*:

$$I_T = I_1 + n\, I_2$$

The ordinary flexure formula then gives the stress in material 1; for material 2, *n* times stiffer, multiply by *n*:

$$\sigma_{x1} = -\frac{M\, y}{I_T} \tag{9}$$

$$\sigma_{x2} = -n\, \frac{M\, y}{I_T} \tag{10}$$

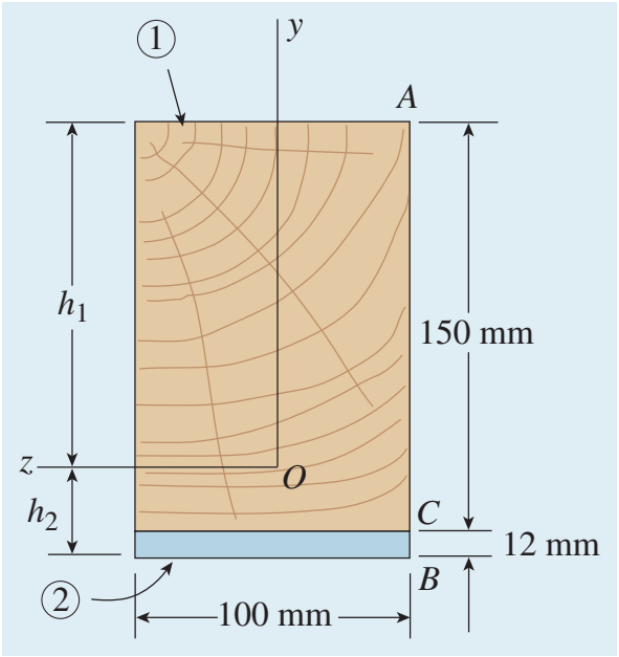
Lecture 5

Composite Beams — worked example

Example 1 Wood and steel beam

Given

- Wood 100 mm × 150 mm, steel plate 100 mm × 12 mm.
- $E_1 = 10.5$ GPa, $E_2 = 210$ GPa.
- $M = 6$ kN·m, sagging moment



Find

The largest stresses in the wood and in the steel.

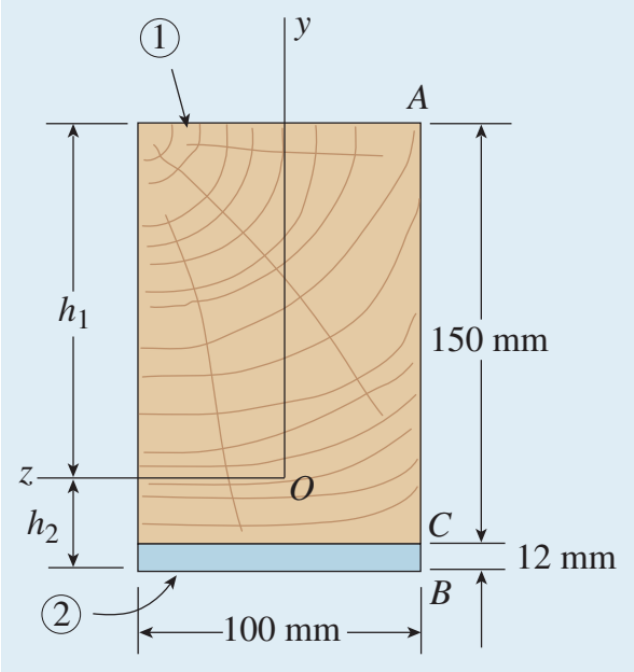
This is textbook Example 6-3.

Example 1 Method 1: Direct solution

Assume measure h_1 downwards from the top surface to the neutral axis.

Wood: $A_1 = 100 \times 150 = 15\,000$ mm², with its centroid 75 mm below the top.

Steel: $A_2 = 100 \times 12 = 1\,200$ mm², with its centroid 150 + 6 = 156 mm below the top.



$$\bar{y}_1 = h_1 - 75, \qquad \bar{y}_2 = h_1 - 156$$

$\bar{y}_1$, $\bar{y}_2$ = distance from the axis to the centroid of each part.

Example 1 Method 1: Direct solution

Each part is a rectangle, and for a rectangle $\int y\, dA = A\, \bar{y}$ with $\bar{y}$ measured from the axis:

$$E_1 A_1 \bar{y}_1 + E_2 A_2 \bar{y}_2 = 0$$

$$10.5(15\,000)(h_1 - 75) + 210(1\,200)(h_1 - 156) = 0$$

$$1.575 \times 10^8 (h_1 - 75) + 2.52 \times 10^8 (h_1 - 156) = 0$$

$$4.095 \times 10^8\, h_1 = 5.1125 \times 10^{10}, \qquad h_1 = 124.8\ \text{mm}$$

$$h_2 = 162\ \text{mm} - h_1 = 37.2\ \text{mm}$$

Example 1 Method 1: Direct solution

Moments of inertia. The moments of inertia areas *A*1 and *A*2 **with respect to the neutral axis** can be found by using the parallel-axis theorem

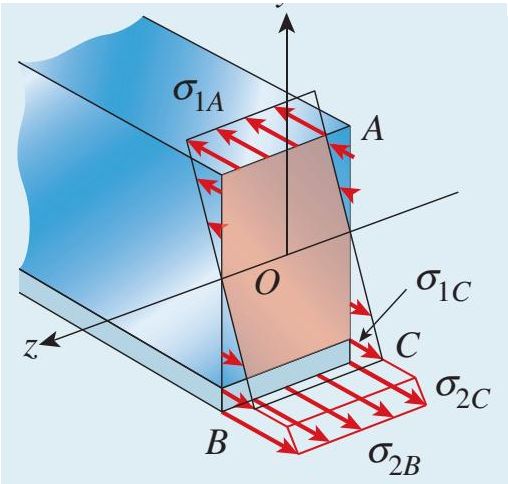
$$I_1 = \frac{1}{12}(100)(150)^3 + (100)(150)(h_1 - 75)^2 = 65.3 \times 10^6 \text{ mm}^4$$

$$I_2 = \frac{1}{12}(100)(12)^3 + (100)(12)(h_2 - 6)^2 = 1.18 \times 10^6 \text{ mm}^4$$

$h_2 = 162 - 124.8 = 37.2$ mm is the depth left below the axis, and 6 mm is half the plate. Each term has two parts: the second moment about the part's own centroid, plus its area times the square of the offset.

Example 1 Method 1: Direct solution

Put the numbers into the flexure formula for the composite beam:



$$\sigma_{1A} = -\frac{Mh_1E_1}{E_1I_1 + E_2I_2}$$

$$= -\frac{(6\ \text{kN}\cdot\text{m})(124.8\ \text{mm})(10.5\ \text{GPa})}{(10.5\ \text{GPa})(65.33 \times 10^6\ \text{mm}^4) + (210\ \text{GPa})(1.18 \times 10^6\ \text{mm}^4)}$$

$$= -8.42\ \text{MPa}$$

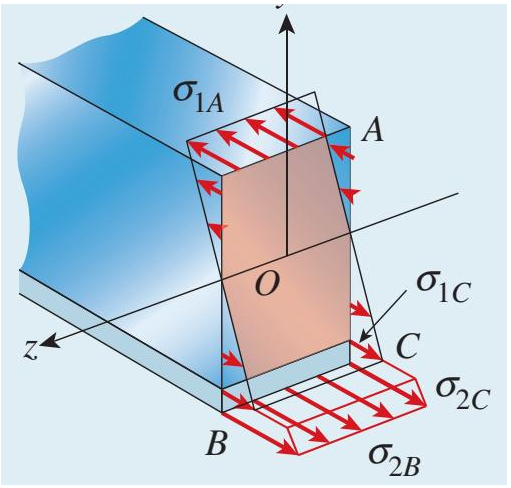
$$\sigma_{1C} = -\frac{(6\ \text{kN}\cdot\text{m})(-25.2\ \text{mm})(10.5\ \text{GPa})}{(10.5\ \text{GPa})(65.33 \times 10^6\ \text{mm}^4) + (210\ \text{GPa})(1.18 \times 10^6\ \text{mm}^4)}$$

$$= 1.7\ \text{MPa}$$

A is the top of the beam, so it is in compression. C is the bonded surface, so it is in tension. These two are the largest stresses in the wood.

Example 1 Method 1: Direct solution

Put the numbers into the flexure formula for the composite beam:



$$\sigma_{2B} = -\frac{M(-h_2)E_2}{E_1 I_1 + E_2 I_2}$$

$$= -\frac{(6\ \text{kN}\cdot\text{m})(-37.2\ \text{mm})(210\ \text{GPa})}{(10.5\ \text{GPa})(65.33 \times 10^6\ \text{mm}^4) + (210\ \text{GPa})(1.18 \times 10^6\ \text{mm}^4)}$$

$$= 50.2\ \text{MPa}$$

$$\sigma_{2C} = -\frac{(6\ \text{kN}\cdot\text{m})(-25.2\ \text{mm})(210\ \text{GPa})}{(10.5\ \text{GPa})(65.33 \times 10^6\ \text{mm}^4) + (210\ \text{GPa})(1.18 \times 10^6\ \text{mm}^4)}$$

$$= 34\ \text{MPa}$$

Note: At the contact plane the ratio of the stress in the steel to the stress in the wood is sam as the modulus ratio E_2/E_1

$$\sigma_{2C}/\sigma_{1C} = 34\ \text{MPa}/1.7\ \text{MPa} = 20$$

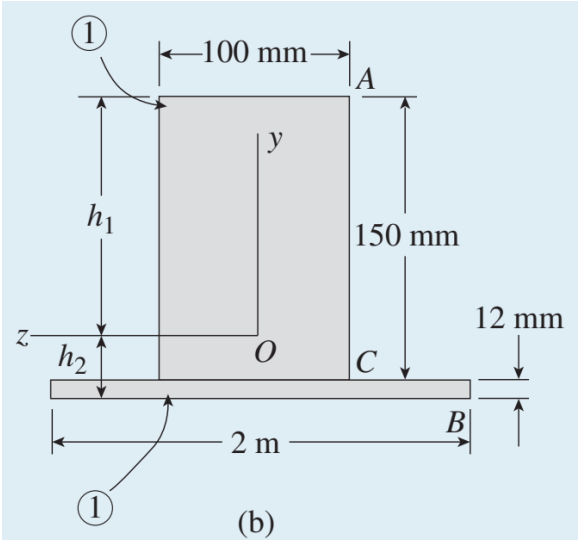
Example 1 Method 2: Transformed Section

The same beam by the other method.
Widen the plate by $n = E_2/E_1 = 20$:

The breadth of bottom steel plate needs to be enlarged by 20 times in transformed section method.

The transformed section is one material, so the axis is its centroid:

The transformed section.



$$h_1 = \frac{\Sigma y_i A_i}{\Sigma A_i} = \frac{(75\text{ mm})(100\text{ mm})(150\text{ mm}) + (156\text{ mm})(2000\text{ mm})(12\text{ mm})}{(100\text{ mm})(150\text{ mm}) + (2000\text{ mm})(12\text{ mm})}$$

$$= \frac{4869 \times 10^3\text{ mm}^3}{39 \times 10^3\text{ mm}^2} = 124.8\text{ mm}$$

$$h_2 = 162\text{ mm} - h_1 = 37.2\text{ mm}$$

The same axis, found without calculating the integral process.

Example 1 Method 2: Transformed Section

Moment of inertia of the transformed section using the parralel-axis theorem:

$$I_T = \frac{1}{12}(100\text{ mm})(150\text{ mm})^3 + (100\text{ mm})(150\text{ mm})(h_1 - 75\text{ mm})^2$$

$$+ \frac{1}{12}(2000\text{ mm})(12\text{ mm})^3 + (2000\text{ mm})(12\text{ mm})(h_2 - 6\text{ mm})^2$$

$$= 65.3 \times 10^6\text{ mm}^4 + 23.7 \times 10^6\text{ mm}^4 = 89.0 \times 10^6\text{ mm}^4$$

Normal stress in the wood:

$$\sigma_{1A} = -\frac{My}{I_T} = -\frac{(6 \times 10^6\text{ N}\cdot\text{mm})(124.8\text{ mm})}{89 \times 10^6\text{ mm}^4} = -8.42\text{ MPa}$$

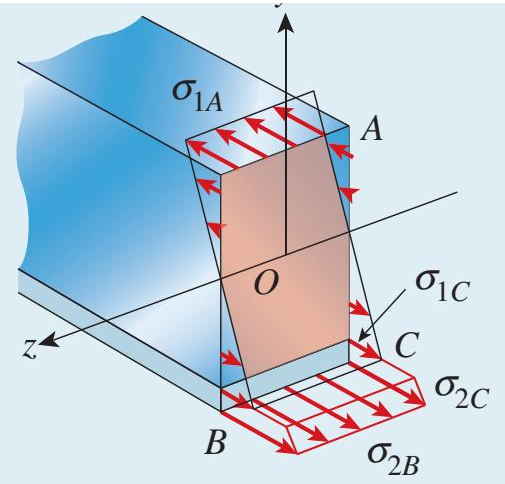
$$\sigma_{1C} = -\frac{My}{I_T} = -\frac{(6 \times 10^6\text{ N}\cdot\text{mm})(-25.2\text{ mm})}{89 \times 10^6\text{ mm}^4} = 1.13\text{ MPa}$$

Example 1 Method 2: Transformed Section

Normal stress in the steel plate part:

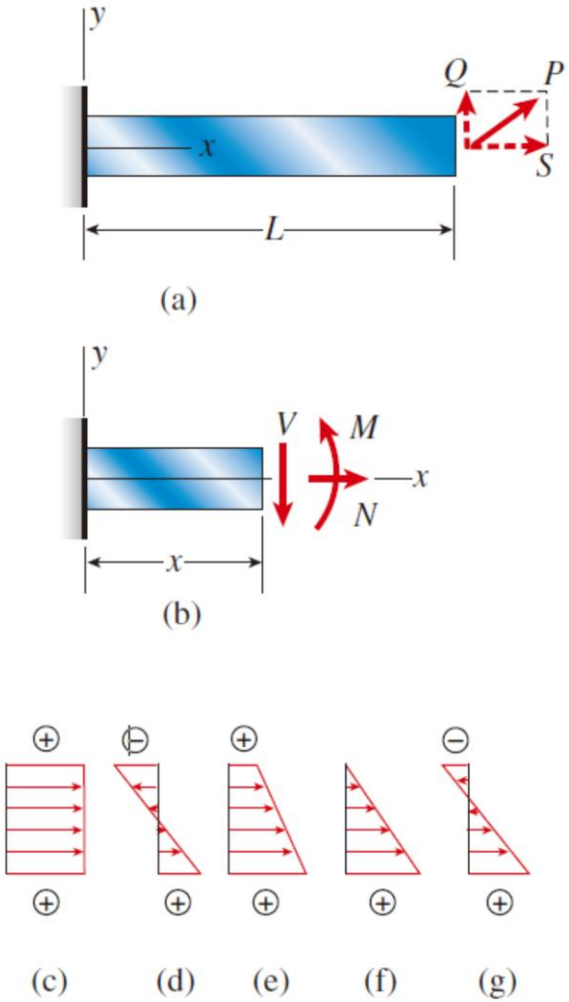
$$\sigma_{2B} = -\frac{My}{I_T} n = -\frac{(6 \times 10^6 \text{ N} \cdot \text{mm})(-37.2 \text{ mm})}{89.0 \times 10^6 \text{ mm}^4} (20) = 50.2 \text{ MPa}$$

$$\sigma_{2C} = -\frac{My}{I_T} n = -\frac{(6 \times 10^6 \text{ N} \cdot \text{mm})(-25.2 \text{ mm})}{89.0 \times 10^6 \text{ mm}^4} (20) = 34 \text{ MPa}$$



Note: At the interface, design calculation is needed to ensure the adhesive or connectors can transfer the interface shear strength between the materials induced by shear force in practical engineering problems.

Beams with axial loads



- An axial force N and a bending moment M act together.
- N alone gives a uniform stress N/A.
- M alone gives a linear stress $-My / I$.
- Add the two to get the final stress.

$$\sigma = \frac{N}{A} - \frac{M\,y}{I} \qquad (11)$$

N = axial force; A = area; M = moment; y = distance from the centroidal axis; I = second moment of area.

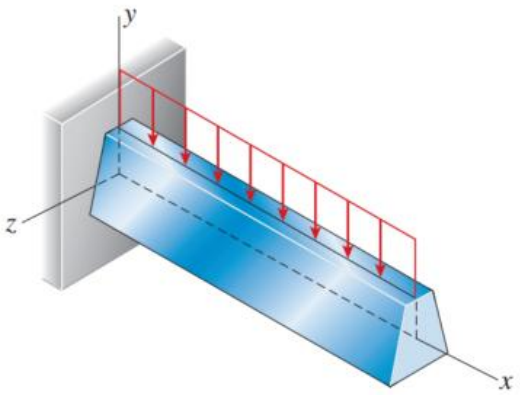
Lecture 5

Doubly Symmetric Beams under Inclined Loads

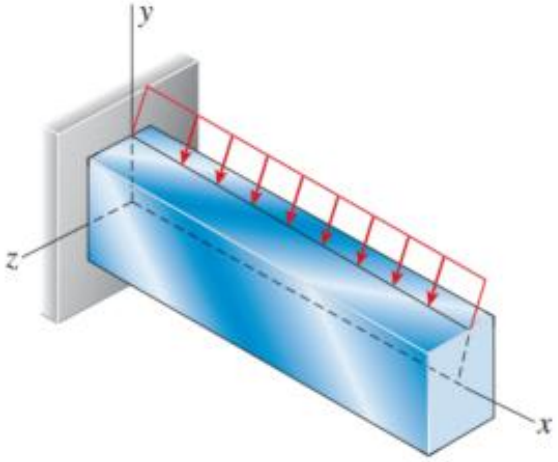
Loads outside a plane of symmetry

- In previous discussions, we assumed beams have a longitudinal plane of symmetry and the loads act in the same plane.
- In this section, assume loads have a component across the section, not only in the plane of bending. Assume both *xy* and *xz* planes are planes of symmetry. Assume

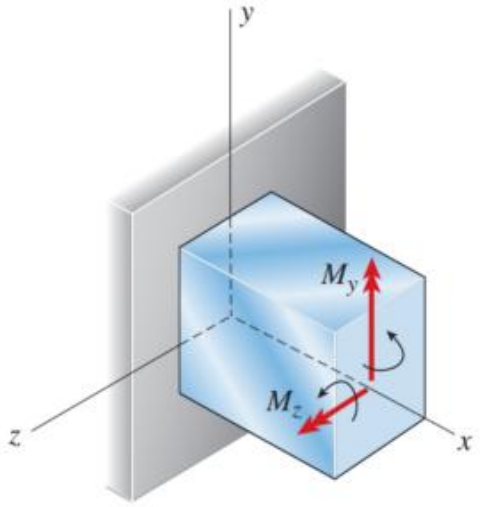
Beam with a lateral load acting in a plane of symmetry



Doubly symmetric beam with an inclined load



Sign conventions for bending moments M_y and M_z

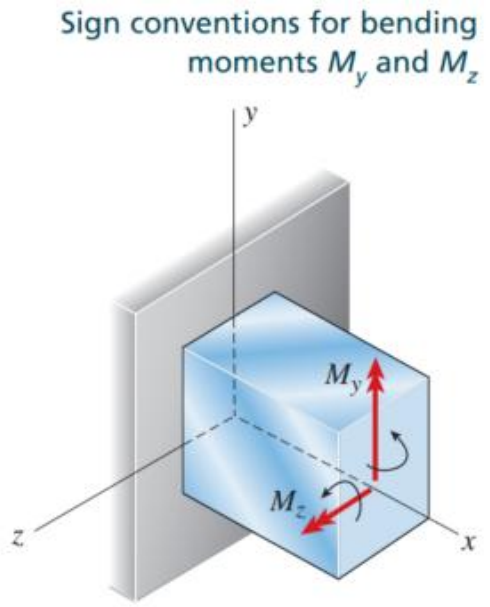


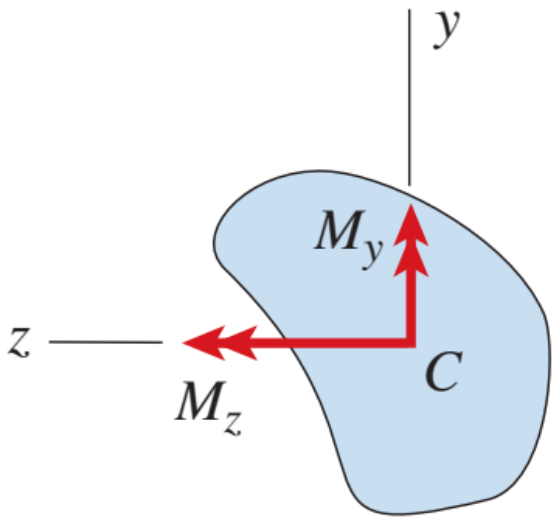
Sign convention and notation

The y axis points up and the z axis points to the left, both through the centroid.

The moment is a vector in the y-z plane; M_z and M_y are its components along the positive axes.

With that convention a positive M_z compresses where y is positive, and a positive M_y pulls where z is positive.





Superposition of the two bendings

A moment about the *z* axis alone gives the flexure formula you already know:

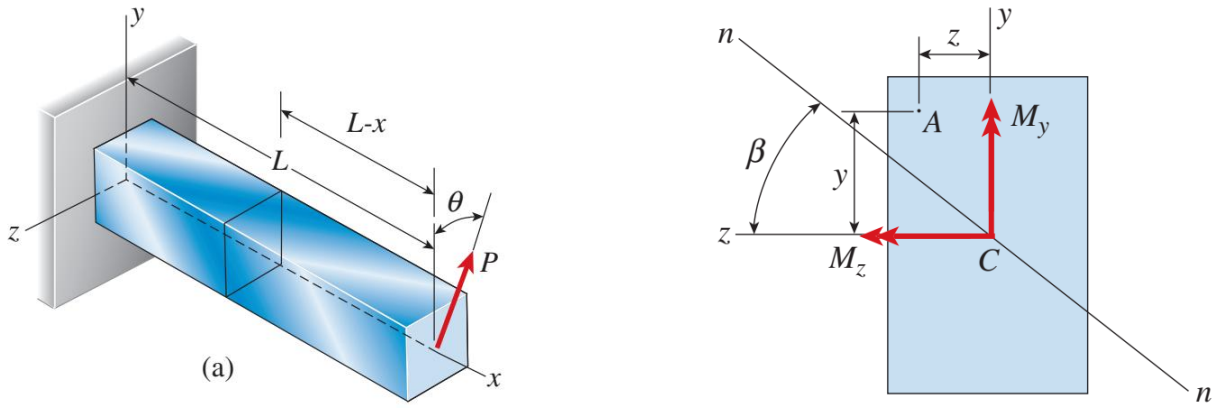
$$\sigma = -\frac{M_z\, y}{I_z}$$

A moment about the *y* axis alone bends the beam the other way, so the stress then depends on *z*:

$$\sigma = +\frac{M_y\, z}{I_y}$$

Adding the two gives the stress for a moment acting in any direction:

Beams under inclined loads



y and *z* are the axes of symmetry, so they are the principal axes. Each moment component then bends the beam in its own plane, and the two stresses add:

$$\sigma_x = \frac{M_y\, z}{I_y} - \frac{M_z\, y}{I_z} \qquad (12)$$

M_y, M_z = moment components; y, z = coordinates of the point.

Neutral axis — step 1: zero stress

The neutral axis is the line along which the stress is zero.

Putting the working formula equal to zero gives:

$$\frac{M_y}{I_y}\, z \;-\; \frac{M_z}{I_z}\, y \;=\; 0 \qquad (13)$$

This is a straight line through the centroid of the section.

Neutral axis — step 2: components

The load P can be resolved into components Pcosθ in the y direction and Psinθ in the negative z direction. The bending moments acting on a cross section at distance x from fixed support are:

$$M_y = (P \sin \theta)(L - x) \qquad M_z = (P \cos \theta)(L - x) \tag{14}$$

Where L=length of the beam

$$\frac{M_y}{M_z} = \tan \theta$$

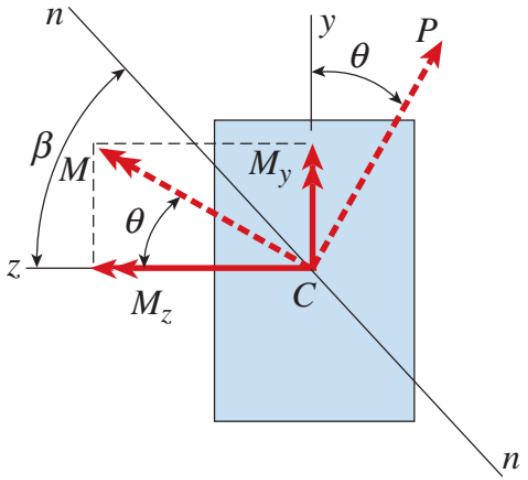
θ = angle between the moment vector M and the z axis.

Neutral axis — step 3: the angle

Substitute the components into the neutral-axis equation. M is a common factor and cancels. If β is the angle of the neutral axis from the z axis:

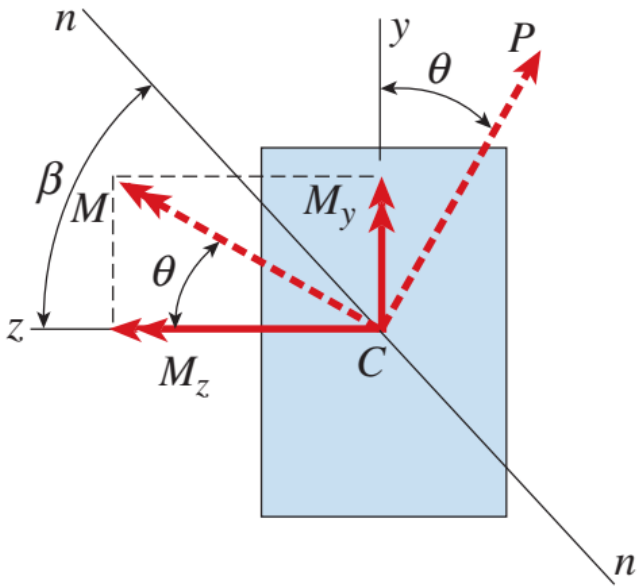
$$\tan \beta = \frac{M_y I_z}{M_z I_y} = \frac{I_z}{I_y} \tan \theta \qquad (15)$$

β = angle of the neutral axis. In general β is not equal to θ.



What the neutral-axis result means

- β equals θ only if $I_y = I_z$, or if M acts along one of the two principal axes.
- When I_z is much larger than I_y, a small θ gives a large β.



In this figure:

Z axis is strong (major) axis

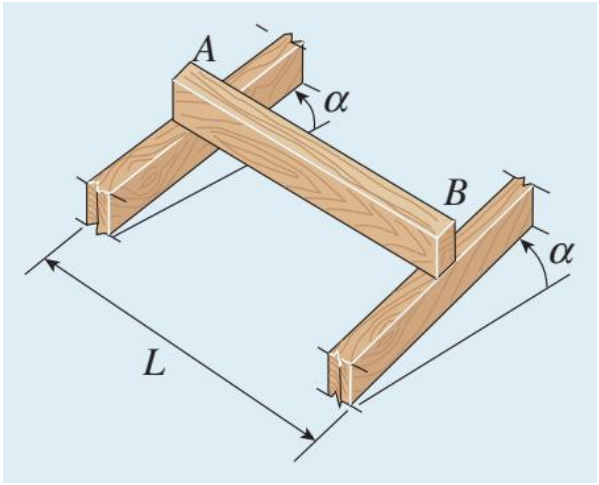
Y axis is weak (minor) axis

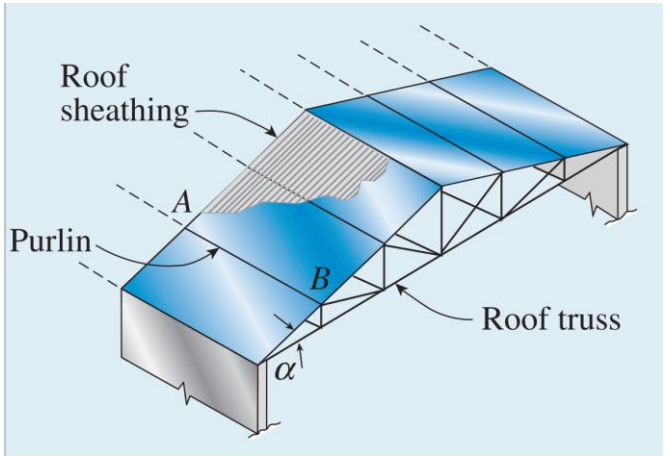
- The largest stress is at whichever point lies farthest from the neutral axis. We can also calculate all stresses at four corners to find the maximum stresses

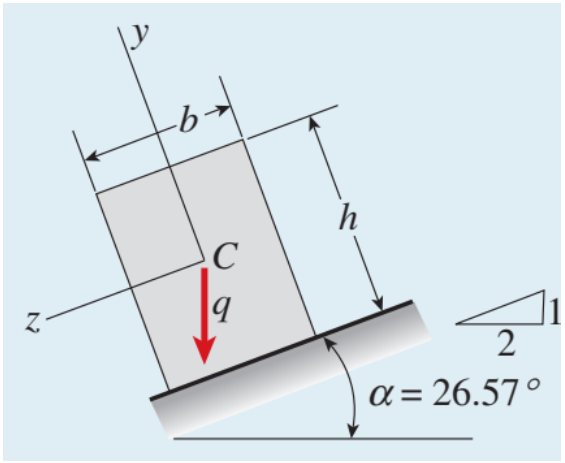
Example 2 A wood roof purlin

Given

- $b = 100$ mm, $h = 150$ mm, $L = 1.6$ m.
- The truss chords slope 1 on 2, so $\alpha = 26.57^\circ$
- Vertical load $q = 3.0$ kN/m through the centroid.





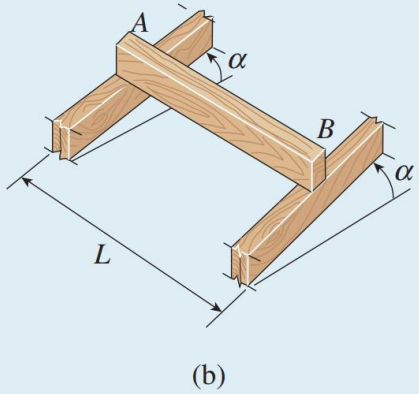


Find

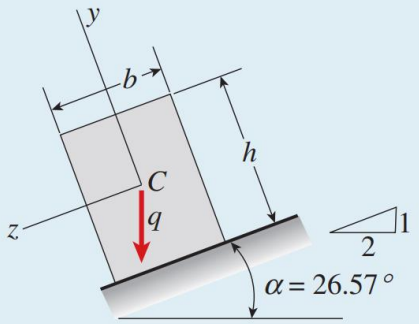
Determine the maximum tensile and compressive stresses in the beam and locate the neutral axis. Textbook Example 6-6.

Example 2 Step 1: resolve the load

The load is vertical, so it is split between the two axes of the tilted section. With α measured from the y axis:



(b)



(c)

$$q_y = q\cos\alpha, \qquad q_z = q\sin\alpha \tag{16}$$

$$M_y = \frac{q_z L^2}{8} = \frac{qL^2\sin\alpha}{8} \qquad M_z = \frac{q_y L^2}{8} = \frac{qL^2\cos\alpha}{8}$$

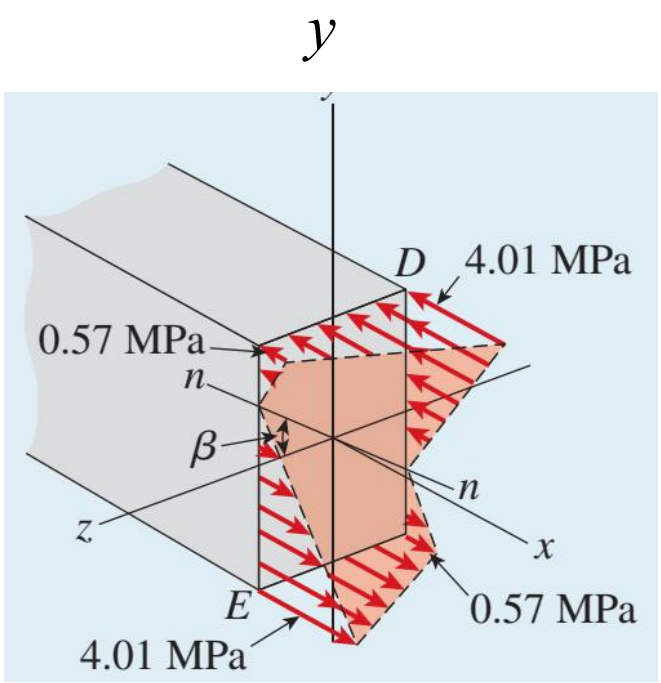
$$I_y = \frac{hb^3}{12} \qquad I_z = \frac{bh^3}{12}$$

Example 2 Step 2: the neutral axis

M is horizontal and *z* is perpendicular to the slope, so the angle from *z* to *M* is the roof slope itself, $\theta = \alpha$:

$$\tan\beta = \frac{I_z}{I_y}\tan\theta = \frac{150^2}{100^2}\tan 26.57^\circ = 1.125$$

$$\beta = 48.4^\circ$$



$\beta \neq \alpha$, because height and breadth are different

Example 2 Step 3: the corner stresses

Put the two moments into the working formula at the two corners that the neutral axis shows to be critical:

$$\sigma_x = \frac{M_y z}{I_y} - \frac{M_z y}{I_z} = \frac{qL^2 \sin\alpha}{8hb^3/12} z - \frac{qL^2 \cos\alpha}{8bh^3/12} y$$

$$= \frac{3qL^3}{2bh}\left(\frac{\sin\alpha}{b^2} z - \frac{\cos\alpha}{h^2} y\right)$$

$$I_z = \frac{100(150)^3}{12} = 28.13 \times 10^6, \qquad I_y = \frac{150(100)^3}{12} = 12.50 \times 10^6 \text{ mm}^4$$

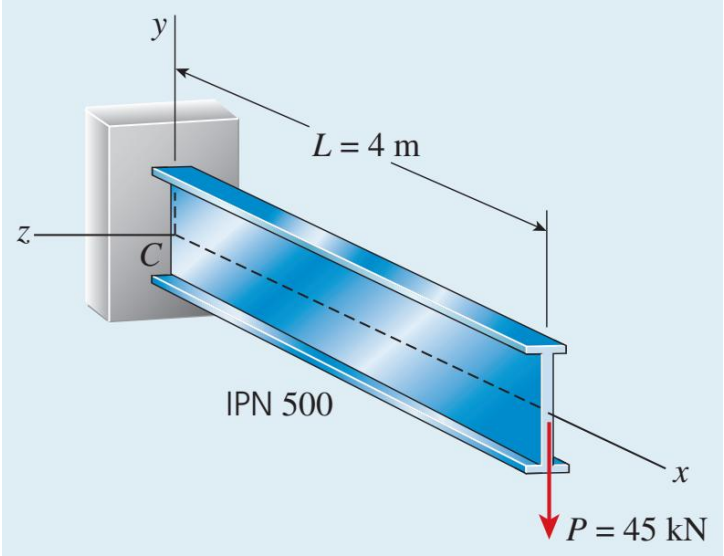
$$\sigma_D = -4.01 \text{ MPa}, \qquad \sigma_E = +4.01 \text{ MPa}$$

The two corners D and *E* are the farthest from the neutral axis, so they carry the largest stresses; the answer also follows from resolving the load into components and adding the two bendings.

Example 3 A slightly tilted cantilever

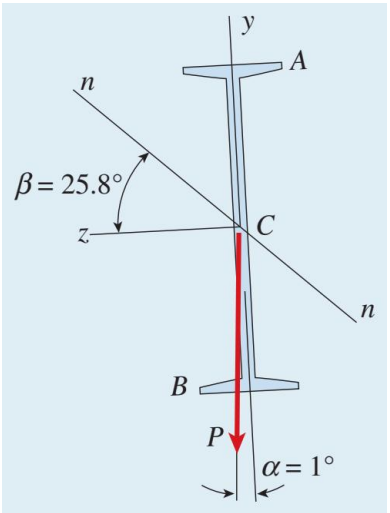
Given

- IPN 500 section cantilever beam $L = 4$ m. • $h = 500$ mm, $b = 185$ mm.
- $P = 45$ kN vertical at the end.
- A narrow section, so I_y is small.



Find

The largest stress, first assume the beam is vertical and then assume the beam is tilted by a small angle of 1 degree. Textbook Example 6-5.



Example 3 Step 1: the beam aligned with the load

With the *y* axis vertical the load bends the beam about *z* alone. The section table gives (this need to be searched from design manuals or website)

$$I_z = 687.4 \times 10^6 \text{ mm}^4, \qquad I_y = 24.8 \times 10^6 \text{ mm}^4$$

The moment at the support and the largest stress are then:

$$M = P\,L = (45)(4) = 180 \text{ kN} \cdot \text{m} \tag{17}$$

$$\sigma_{max} = \frac{M\,(h/2)}{I_z} = \frac{(180 \times 10^6)(250)}{687.4 \times 10^6} = 65.5 \text{ MPa}$$

Tensile at the top, compressive at the bottom.

Example 3 Step 2: a tilt of one degree

Tilt the beam by $\alpha = 1^\circ$. The load now has a component along z, so both moments act at the support:

$$M_z = -(P\cos\alpha)L = -(45)(0.9998)(4) = -180\ \text{kN}\cdot\text{m} \qquad (18)$$

$$M_y = -(P\sin\alpha)L = -(45)(0.01745)(4) = -3.14\ \text{kN}\cdot\text{m}$$

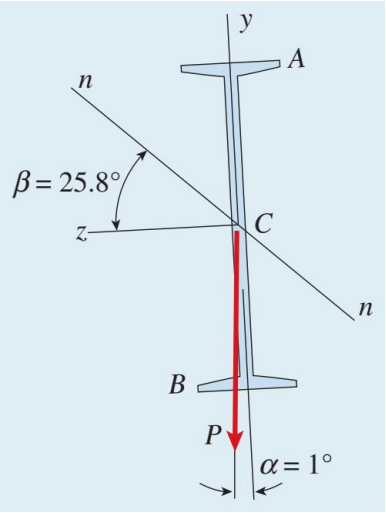
The moment vector is only 1° from the z axis, but the inertia ratio magnifies the angle:

$$\tan\beta = \frac{I_z}{I_y}\tan\alpha = 27.7 \times 0.01746 = 0.484, \qquad \beta = 25.8^\circ$$

Example 3 Step 3: the corner stresses

The corners farthest from nn are A at the top and B at the bottom. At A, $z_A = -92.5$ mm and $y_A = +250$ mm:

$$\sigma_A = \frac{M_y z_A}{I_y} - \frac{M_z y_A}{I_z} = 11.7 + 65.5 = +77.2 \text{ MPa}$$



$$\sigma_B = -77.2 \text{ MPa}$$

Both stresses are 18 % larger than the 65.5 MPa of the aligned beam, and the beam also bends sideways (lateral deformation). A narrow steel beam needs lateral support to avoid such loading.

Formula summary (1): composite beams

$$\int_{A_1} E_1\, y\, dA + \int_{A_2} E_2\, y\, dA = 0$$

locates the neutral axis

$$\kappa = \frac{M}{E_1 I_1 + E_2 I_2}$$

curvature of the composite beam

$$\sigma_{x1} = -E_1 \kappa\, y, \qquad \sigma_{x2} = -E_2 \kappa\, y$$

stress in each material

$$n = \frac{E_2}{E_1}, \qquad b_2{}^{*} = n\, b_2$$

the transformed section

$$\sigma_{x2} = n\, \sigma_{x1}{}^{tr}$$

back to the real stress in material 2

Formula summary (2): inclined loads

$$\sigma_x = \frac{M_y\, z}{I_y} - \frac{M_z\, y}{I_z}$$

the working formula for the whole section

$$M_z = M\cos\theta\,, \qquad M_y = M\sin\theta$$

the two components of M about the principal axes

$$\tan\beta = \frac{I_z}{I_y}\tan\theta$$

β from z to the neutral axis; it equals θ only if $I_y = I_z$

Summary — composite beams

- Strains stay linear; the neutral axis is not at the geometric centroid.
- $\kappa = M / (E_1 I_1 + E_2 I_2)$.
- Widen material 2 by $n = E_2/E_1$ to build the transformed section.
- Multiply the transformed stress by n to get the stress in material 2.

Summary — inclined loads

- Resolve the moment about the principal axes: $M_z = M\cos\theta$, $M_y = M\sin\theta$.
- $\sigma_x = M_y\, z / I_y - M_z\, y / I_z$
- β follows from $\tan\beta = (I_z/I_y)\tan\theta$; it equals θ only if $I_y = I_z$.
- The largest stress is at the point farthest from the neutral axis.

Common mistakes

- Using the transformed centroid instead of the geometric one.
- Forgetting that the stress ratio at a bonded surface is E_2/E_1.
- Assuming the neutral axis is perpendicular to the plane of loading.
- Looking for the largest stress only at the top and bottom.
- Dropping the small moment component about the weak axis.
- Mixing kN·m with mm^4 without converting to N·mm first.

The End of Lecture 5

Next: shear stress and the shear centre